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Surjection: at least 1 arrow

$$S \subseteq A$$

$$\forall y \in B: \exists x \in A: f(x) = y$$

1.)

Is a surjection: $\forall y \in B: \exists x \in A: f(x) = y$

$$\text{var } a, a \in B$$

$$\exists x \in A: f(x) = a$$

Pick an x with $x \in A$ and $f(x) = a$

$$\neg (a \in f(S))$$

{prop of image}

$$\neg (\exists b \in S: f(b) = a)$$

pick an b with $b \in S$ and $f(b) = a$

$$b \in S \wedge S \subseteq A$$

$$b \in A$$

$$\neg (\exists b \in A: f(b) = a)$$

False

$$a \in f(A \setminus S)$$

$$a \in f(S) \vee a \in f(A \setminus S)$$

{prop of $\cup$ }

$$a \in f(S) \cup f(A \setminus S)$$

$$\forall a \in B: a \in f(S) \cup f(A \setminus S)$$

{prop of $\subseteq$ }

$$B \subseteq f(S) \cup f(A \setminus S)$$